

Lecture 03: Relationships between 2 variables

Time plots

Relationships between two
quantitative variables

Looking at relationships
visually: Scatterplots

Exploratory analysis using
scatterplots

Assessing a relationship
between two variables with a
number: Pearson's
correlation

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Recap of chapters 1 and 2

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Mostly looking at a single variable:

- ▶ Graphs to explore the distribution of single variables (histograms, bar charts)
- ▶ Summary numbers to describe our distributions:
 - ▶ Measures of central tendency (mean, median)
 - ▶ Measures of spread (standard deviation, IQR)

Learning objectives for today

- ▶ Time plots to examine what happens to a variable over time
 - ▶ using `geom_point()`
 - ▶ using `geom_line()`
- ▶ Explore the relationship between two quantitative variables
 - ▶ Directionality
 - ▶ Association vs causation
- ▶ Make scatter plots to look at relationships visually
 - ▶ using `geom_point()`
- ▶ Use the correlation coefficient to quantify the strength of linear relationships
 - ▶ calculate correlations using `cor()`

Time plots

Relationships between two quantitative variables

Looking at relationships visually: Scatterplots

Exploratory analysis using scatterplots

Assessing a relationship between two variables with a number: Pearson's correlation

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Time plots

Visualize quantitative variables over time using time plots

Time plots

Relationships between two quantitative variables

Looking at relationships visually: Scatterplots

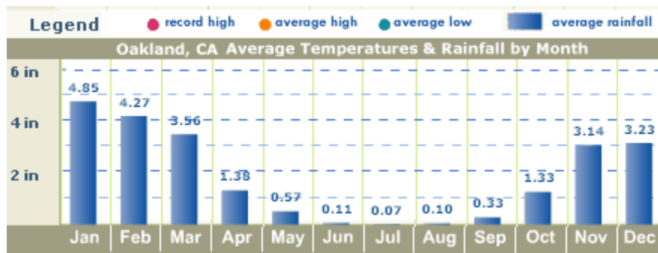
Exploratory analysis using scatterplots

Assessing a relationship between two variables with a number: Pearson's correlation

- ▶ **Time plots** are a specific subset of plots where the x variable is time.
- ▶ Unlike the previous plots, the time plot shows a relationship between two variables:
 - i) a quantitative variable
 - ii) time
- ▶ Often times, these plots can be used to look for cycles (e.g., seasonal patterns that recur each year) or trends (e.g., overall increases or decreases seen over time).

Time plot

► from [See California.com](http://SeeCalifornia.com), January 2021:



Time plots

Relationships between two quantitative variables

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Assessing a relationship between two variables with a number: Pearson's correlation

Life expectancy for White men in California

Time plots

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Make a scatter plot of the life expectancy for White men in California over time.

Since the dataset contains 39 states across two genders and two races, first use a function to subset the data to contain only White men in California.

Which function from last lecture do we need?

► `mutate()`, `select()`, `filter()`, `rename()`, or `arrange()`?

dplyr's filter() to select a subset of rows

Time plots

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```
wm_cali <- le_data %>% filter(state == "California",  
                               sex == "Male",  
                               race == "white")
```

#this is equivalent:

```
wm_cali <- le_data %>% filter(state == "California" & sex == "Male" & race ==
```

Here we use `geom_point` to make a graph with dots

```
ggplot(data = wm_cali, aes(x = year, y = LE)) +  
  geom_point() +  
  labs(title = "Life expectancy in white men in California, 1969-2013",  
        y = "Life expectancy",  
        x = "Year",  
        caption = "Data from Riddell et al. (2018)")
```

Time plots

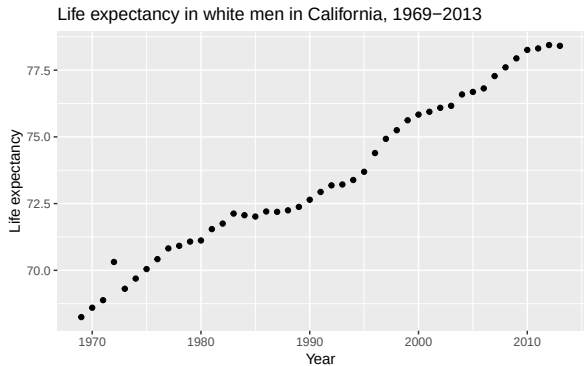
Relationships between two
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Here we use `geom_point` to make a graph with dots



Data from Riddell et al. (2018)

Time plots

Relationships between two quantitative variables

Looking at relationships visually: Scatterplots

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Assessing a relationship between two variables with a number: Pearson's correlation

geom_line() to make a line plot

Time plots

Relationships between two
quantitative variables

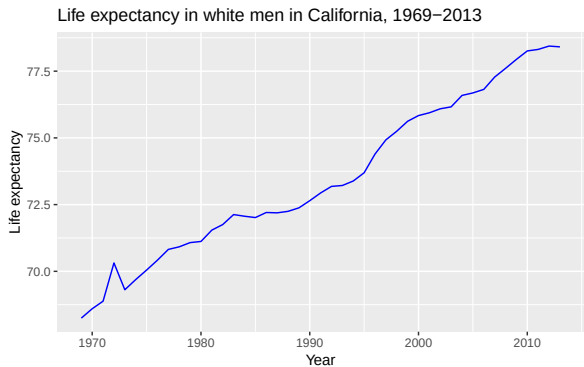
Looking at relationships
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```
ggplot(data = wm_cali, aes(x = year, y = LE)) +  
  geom_line(col = "blue") +  
  labs(title = "Life expectancy in white men in California, 1969-2013",  
        y = "Life expectancy",  
        x = "Year",  
        caption = "Data from Riddell et al. (2018)")
```

geom_line() to make a line plot



Data from Riddell et al. (2018)

Time plots

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Relationships between two quantitative variables

Explanatory (X) and response (Y) variables

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Bi-directional:

- ▶ “X predicts Y”, or “Y predicts X”
- ▶ “X is associated with Y”, or “Y is associated with X”

Unidirectional:

- ▶ “X causes Y”

Which variable is x and which is y ?

Time plots

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In prediction we generally use X to denote the variable we are using to predict the variable of interest (Y)

In causation we generally use X to denote the explanatory (independent) and Y to denote the response (dependent)

Graphically the X variable is on the X (horizontal) axis and the Y variable is the Y (vertical) axis

Which variable is x and which is y?

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1. Each hospital's rate of hospital-acquired infections, and whether the hospital has implemented a hand-washing intervention as part of a cluster randomized trial.
2. The weight in kilograms and height in centimeters of a person
3. Inches of rain in the growing season and the yield of corn in bushels per day
4. A person's leg length and arm length, in centimeters

How to investigate causation?

Time plots

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- ▶ Randomized controlled trials (RCTs) to randomize individuals to different levels
- ▶ Observational study that is *designed* to investigate causation and reduce the risk of bias

Time plots

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Exploratory analysis using
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Assessing a relationship
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- ▶ Scatterplots are a good way to visualize a relationship between two variables
- ▶ When we look at a scatterplot we want to evaluate:
 - ▶ The overall Pattern of the dots
 - ▶ Any notable exceptions to the pattern
 - ▶ Direction (positive or negative)
 - ▶ Form (straight line or curved)
 - ▶ Strength (how closely the points follow a line)
 - ▶ Are there any obvious outliers

Scatterplot Syntax in R

Time plots

Relationships between two
quantitative variables

**Looking at relationships
visually: Scatterplots**

Exploratory analysis using
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Assessing a relationship
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```
name of plot <- ggplot(data = dataset, aes(x = xvariable, y = yvariable)) +  
geom_point(na.rm=TRUE) + theme_minimal(base_size = 15)+  
labs(x = "xlabel", y = "ylabel", title = "Title")
```

Bi-directional relationships ex: systolic and diastolic BP

Read in NHANES dataset

```
nhanes_dataNA <- read_csv("nhanes.csv")
nhanes_data <- nhanes_dataNA[rowSums(is.na(nhanes_dataNA[, 15:18])) == 0, ]
names(nhanes_data)
```

```
## [1] "ridageyr" "agegroup" "gender" "military" "born" "citizen"
## [7] "drinks" "drinkscat" "bmxwt" "bmxht" "bmxbmi" "bmicat"
## [13] "bpxpls" "bpxsy1" "bpxsy2" "sys1d" "sys2d" "bpxdi1"
## [19] "bpxdi2" "dias1d" "dias2d" "bpcat" "chest" "fs1"
## [25] "fs2" "fs3" "lbdhdd" "hdlcat" "highhdl" "hi"
## [31] "asthma" "vwa" "vra" "va" "aspirin" "sleep"
## [37] "is" "hs" "lbdldl" "highldl"
```

Time plots

Relationships between two
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Looking at relationships
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Exploratory analysis using
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Bi-directional relationships ex: systolic and diastolic BP

Time plots

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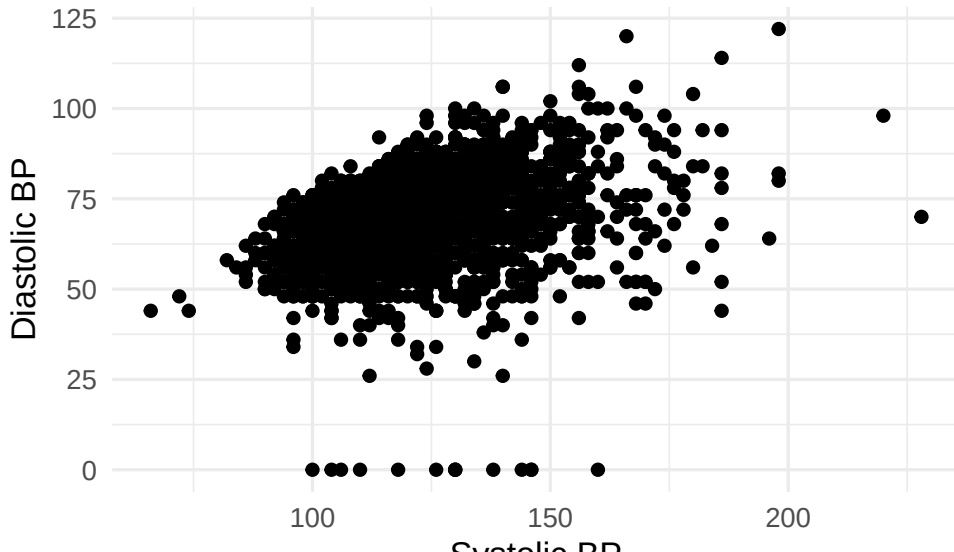
Exploratory analysis using
scatterplots

Assessing a relationship
between two variables with a
number: Pearson's
correlation

```
nhanes_scatter <- ggplot(data = nhanes_data, aes(x = bpxsy1, y = bpxdi1)) +  
  geom_point(na.rm=TRUE) + theme_minimal(base_size = 15)+  
  labs(x = "Systolic BP",  
       y = "Diastolic BP",  
       title = "NHANES Data")
```

Bi-directional relationships ex: systolic and diastolic BP

NHANES Data



Time plots

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Bi-directional relationships ex: systolic and diastolic BP

Time plots

Relationships between two
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**Looking at relationships
visually: Scatterplots**

Exploratory analysis using
scatterplots

Assessing a relationship
between two variables with a
number: Pearson's
correlation

What do we notice from the plot?

- ▶ Is there a visible association?
- ▶ Any notable exceptions to the pattern
- ▶ Direction (positive or negative)
- ▶ Form (straight line or curved)
- ▶ Strength (how closely the points follow a line)
- ▶ Are there any obvious outliers

Bi-directional relationships ex: systolic and diastolic BP

Time plots

Relationships between two
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Exploratory analysis using
scatterplots

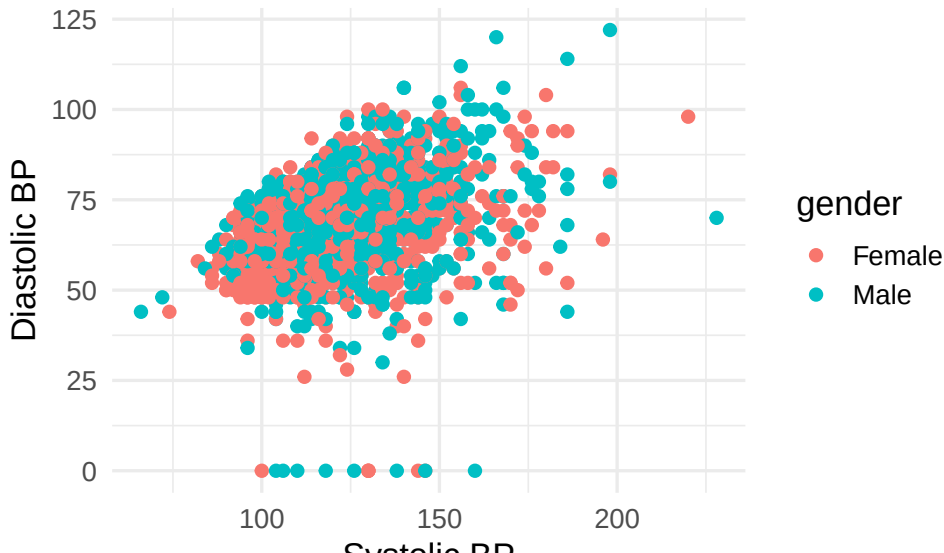
Assessing a relationship
between two variables with a
number: Pearson's
correlation

We can add a third variable to our graph by coloring the dots

```
nhanes_scatter <- ggplot(data = nhanes_data, aes(x = bpxsy1, y = bpxdi1)) +  
  geom_point(aes(col=gender), na.rm=TRUE) + theme_minimal(base_size = 15) +  
  labs(x = "Systolic BP",  
        y = "Diastolic BP",  
        title = "NHANES Data")
```

Bi-directional relationships ex: systolic and diastolic BP

NHANES Data



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Association with a plausible direction

Manatee data set from your textbook:

```
mana_data <- read_csv("Ch03_Manatee-deaths.csv")  
head(mana_data)
```

```
## # A tibble: 6 x 3  
##   year powerboats deaths  
##   <dbl>     <dbl>   <dbl>  
## 1  1977         447     13  
## 2  1987         645     39  
## 3  1997         755     54  
## 4  2007        1027     73  
## 5  1978         460     21  
## 6  1988         675     43
```

Time plots

Relationships between two
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Looking at relationships
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Exploratory analysis using
scatterplots

Assessing a relationship
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Power boats and Manatees

Time plots

Relationships between two
quantitative variables

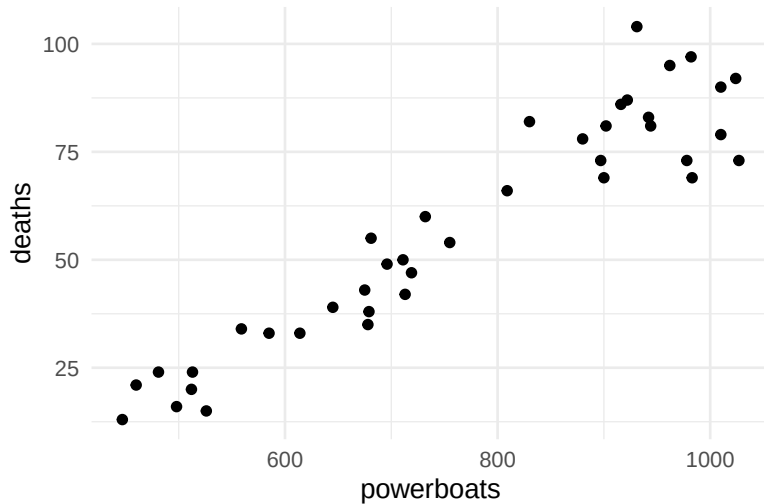
**Looking at relationships
visually: Scatterplots**

Exploratory analysis using
scatterplots

Assessing a relationship
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number: Pearson's
correlation

```
mana_scatter <- ggplot(data = mana_data, aes(x = powerboats, y = deaths)) +  
  geom_point() + theme_minimal(base_size = 15)
```

Power boats and Manatees



Time plots

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Time plots

Relationships between two
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Exploratory analysis using
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What do we notice from the plot?

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- ▶ Are there any obvious outliers

Power boats and Manatees

Time plots

Relationships between two
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Looking at relationships
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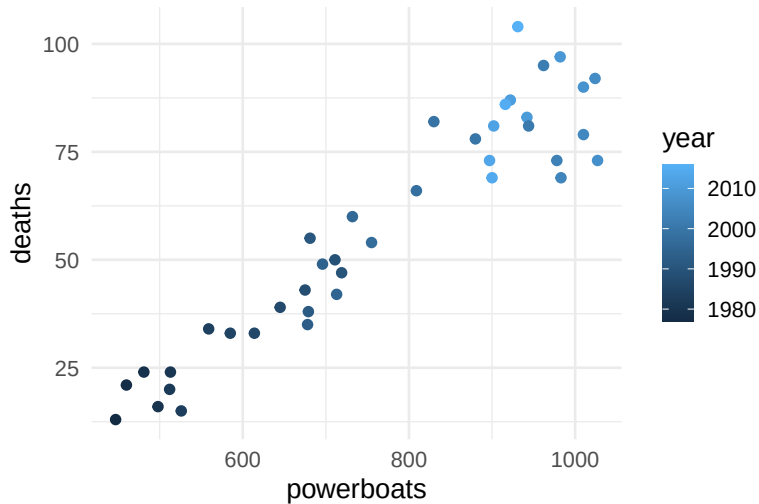
Exploratory analysis using
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Assessing a relationship
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number: Pearson's
correlation

What if we layer in a continuous third variable?

```
mana_scatter <- ggplot(data = mana_data, aes(x = powerboats, y = deaths)) +  
  geom_point(aes(col=year)) + theme_minimal(base_size = 15)
```


Power boats and Manatees



Time plots

Relationships between two
quantitative variables

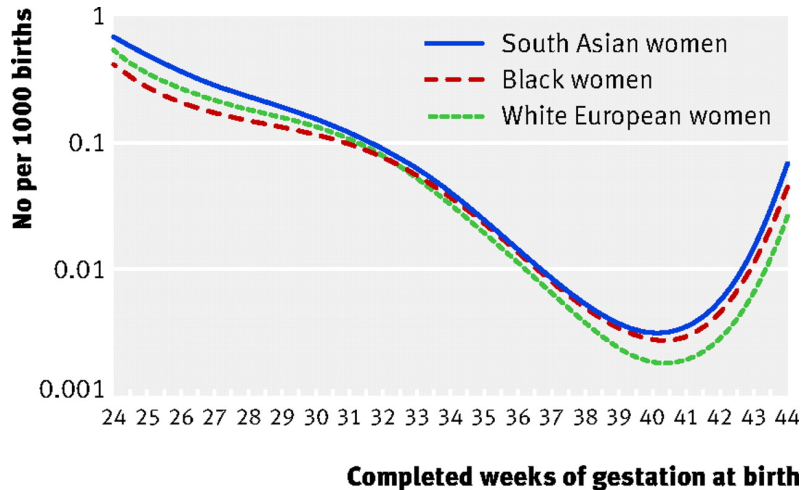
Looking at relationships
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Assessing a relationship
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A non-linear example

Gestational age and perinatal mortality



Source: Balchin et al. BMJ. 2007.

Time plots

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**Exploratory analysis using
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Exploratory analysis using scatterplots

Lean body mass and metabolic rate: Problem and Plan

Lecture 03:
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**Exploratory analysis using
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Problem: Is lean body mass (person's weight after removing the fat) associated with metabolic rate (kilocalories burned in 24 hours)?

Plan: A diet study was conducted on 12 women and 7 men that measured lean body weight and metabolic rate for each individual.

Lean body mass and metabolic rate: DATA

Data: In the textbook

Subject	Sex	Mass (kg)	Rate (Cal)	Subject	Sex	Mass (kg)	Rate (Cal)
1	M	62.0	1792	11	F	40.3	1189
2	M	62.9	1666	12	F	33.1	913
3	F	36.1	995	13	M	51.9	1460
4	F	54.6	1425	14	F	42.4	1124
5	F	48.5	1396	15	F	34.5	1052
6	F	42.0	1418	16	F	51.1	1347
7	M	47.4	1362	17	F	41.2	1204
8	F	50.6	1502	18	M	51.9	1867
9	F	42.0	1256	19	M	46.9	1439
10	M	48.7	1614				

What would the corresponding data frame look like? How many variables would it have? How many rows?

Time plots

Relationships between two
quantitative variables

Looking at relationships
visually: Scatterplots

Exploratory analysis using
scatterplots

Assessing a relationship
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Lean body mass and metabolic rate: DATA

Note: you won't be tested on writing code using `tibble::tribble()`

Do be able to look at the code and recognize that it is creating a data set

```
weight_data <- tibble::tribble(  
  ~subject, ~gender, ~mass, ~rate,  
  1, "M", 62.0, 1792,  
  2, "M", 62.9, 1666,  
  3, "F", 36.1, 995,  
  4, "F", 54.6, 1425,  
  5, "F", 48.5, 1396,  
  6, "F", 42.0, 1418,  
  7, "M", 47.4, 1362,  
  8, "F", 50.6, 1502,  
  9, "F", 42.0, 1256,  
  10, "M", 48.7, 1614,  
  11, "F", 40.3, 1189,  
  12, "F", 33.1, 913,
```

Lean body mass and metabolic rate: Analysis

Time plots

Relationships between two
quantitative variables

Looking at relationships
visually: Scatterplots

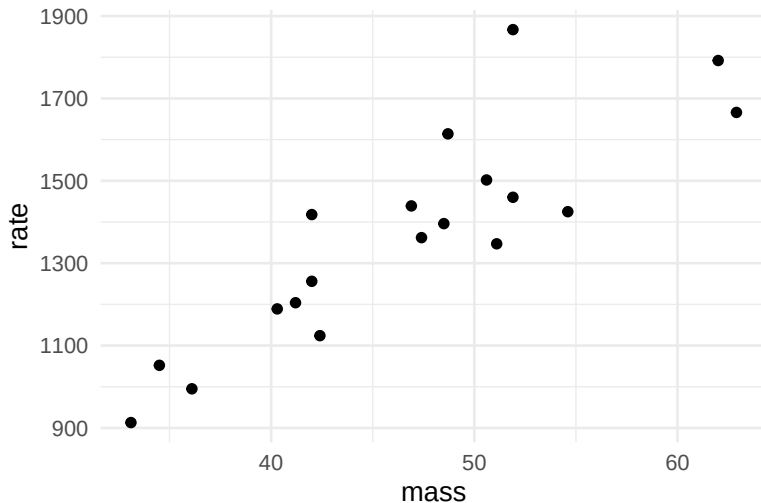
**Exploratory analysis using
scatterplots**

Assessing a relationship
between two variables with a
number: Pearson's
correlation

Exploratory data analysis using scatter plots

```
weight_scatter <- ggplot(weight_data, aes(x = mass, y = rate)) +  
  geom_point() +  
  theme_minimal(base_size = 15)
```

Lean body mass and metabolic rate: Analysis



Time plots

Relationships between two
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**Exploratory analysis using
scatterplots**

Assessing a relationship
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Analysis: Colour the points by gender

Time plots

Relationships between two
quantitative variables

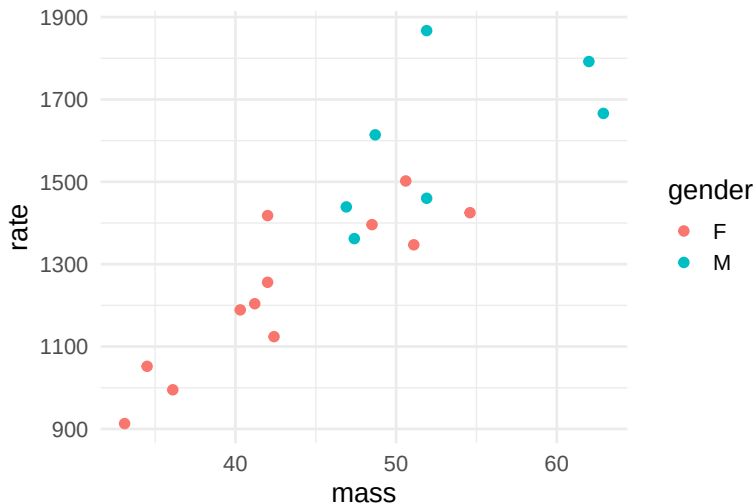
Looking at relationships
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**Exploratory analysis using
scatterplots**

Assessing a relationship
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number: Pearson's
correlation

```
weight_scatter <- ggplot(weight_data, aes(x = mass, y = rate)) +  
  geom_point(aes(col=gender)) +  
  theme_minimal(base_size = 15)
```

Analysis: Colour the points by gender



Time plots

Relationships between two
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**Exploratory analysis using
scatterplots**

Assessing a relationship
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Analysis: Create separate plots for men and women

Time plots

Relationships between two
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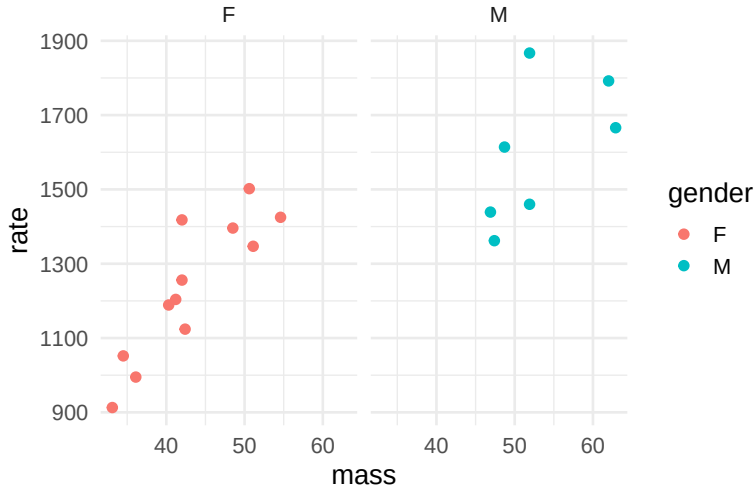
**Exploratory analysis using
scatterplots**

Assessing a relationship
between two variables with a
number: Pearson's
correlation

```
weight_scatter <- ggplot(weight_data, aes(x = mass, y = rate)) +  
  geom_point(aes(col=gender)) +  
  theme_minimal(base_size = 15)+  
  facet_wrap(~ gender)
```

Analysis: Create separate plots for men and women

weight_scatter



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Assessing a relationship between two variables with a number: Pearson's correlation

Pearson's correlation

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Using just our eyes, we can often say something about whether an association between two variables is weak or strong.

But we can also use a numeric value to describe the direction and strength of an association

Pearson's correlation

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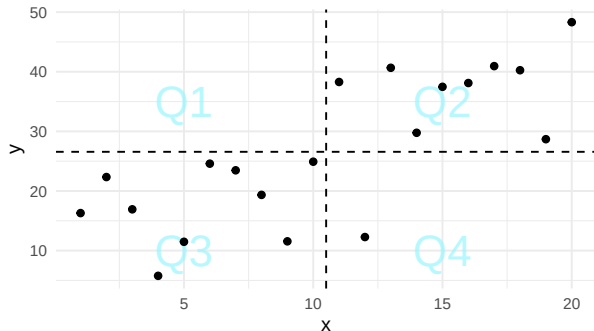
- ▶ For **linear** associations, we can use **Pearson's correlation coefficient** (denoted by r) to **quantify the strength** of a linear relationship between two variables.
- ▶ The correlation between x and y is:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Notice that because we are dividing by the standard deviation the values become unitless

Intuition about correlation: center the variables

- ▶ First compare each value to its own mean: $x_i - \bar{x}$ and $y_i - \bar{y}$.
- ▶ The dashed lines split the plot into four regions.
- ▶ Each point is above or below the average value of each variable.



Time plots

Relationships between two
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Looking at relationships
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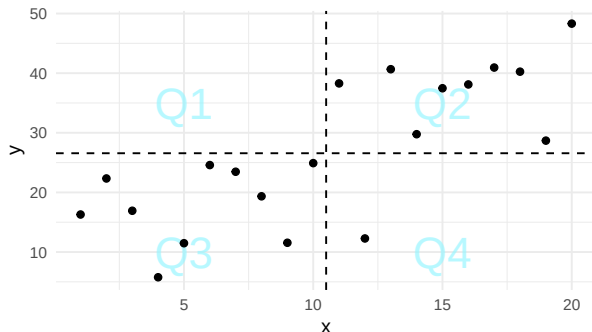
Exploratory analysis using
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Assessing a relationship
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number: Pearson's
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Intuition about correlation: signs

$$r = \frac{1}{n-1} \sum_{i=1}^n z_{x,i} z_{y,i}$$

- ▶ Same sign deviations give a **positive** contribution.
- ▶ Opposite sign deviations give a **negative** contribution.



Time plots

Relationships between two
quantitative variables

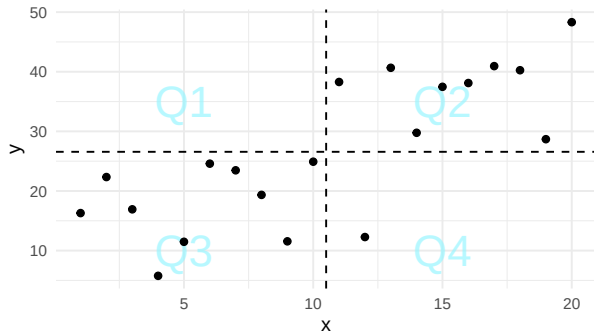
Looking at relationships
visually: Scatterplots

Exploratory analysis using
scatterplots

Assessing a relationship
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number: Pearson's
correlation

Intuition about correlation: direction

- ▶ More points in Q2 and Q3 $\Rightarrow$ positive correlation.
- ▶ More points in Q1 and Q4 $\Rightarrow$ negative correlation.
- ▶ A balance across regions $\Rightarrow$ correlation near 0.



Time plots

Relationships between two
quantitative variables

Looking at relationships
visually: Scatterplots

Exploratory analysis using
scatterplots

Assessing a relationship
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Properties of the correlation coefficient

Time plots

Relationships between two
quantitative variables

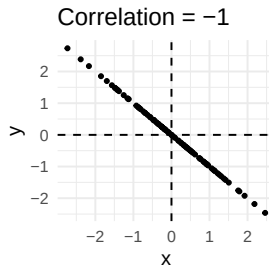
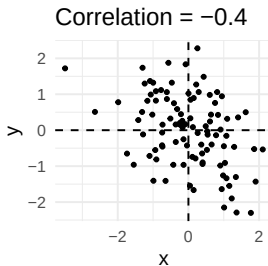
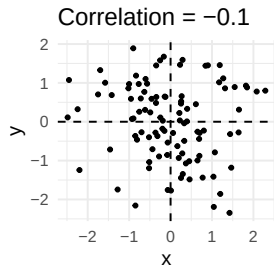
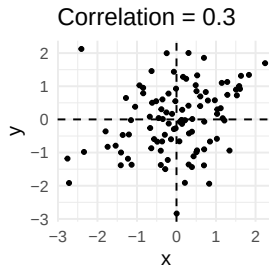
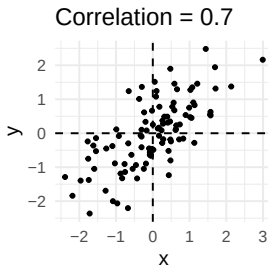
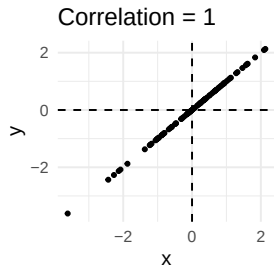
Looking at relationships
visually: Scatterplots

Exploratory analysis using
scatterplots

Assessing a relationship
between two variables with a
number: Pearson's
correlation

- ▶ Always a number between -1 and 1.
 - ▶ -1: A perfect, negative linear association
 - ▶ 1: A perfect, positive linear association
 - ▶ 0: No linear association
- ▶ Is used to measure the association between two *quantitative* variables.
- ▶ Only useful for *linear* associations!

Correlation and direction



Time plots

Relationships between two quantitative variables

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Assessing a relationship between two variables with a number: Pearson's correlation

Syntax: Pearson's correlation using cor()

Time plots

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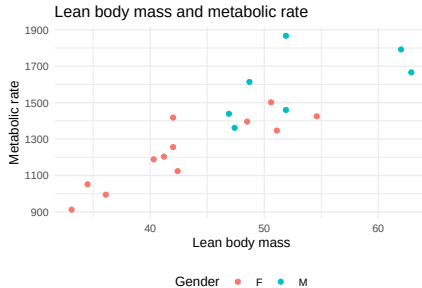
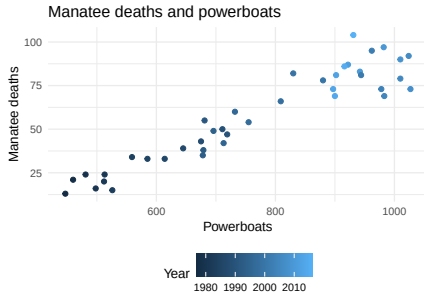
Assessing a relationship
between two variables with a
number: Pearson's
correlation

```
correlation coefficient <- dataset %>%
```

```
summarize(newvar = cor(xvar, yvar))
```

Remember the manatee and weight plots

Before calculating r , predict the sign and rough strength from the plots.



Time plots

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Correlation for manatee data

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Now, calculate the correlations between X and Y for manatees:

```
mana_cor <- mana_data %>%  
  summarize(corr_mana = cor(powerboats, deaths))  
mana_cor
```

```
## # A tibble: 1 x 1  
##   corr_mana  
##   <dbl>  
## 1      0.945
```

Correlation for weight data

Time plots

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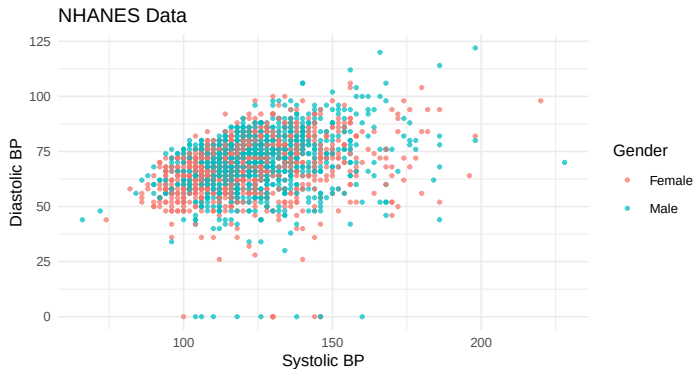
And for the weight data:

```
weight_cor <- weight_data %>%  
  summarize(corr_weight = cor(mass, rate))  
weight_cor
```

```
## # A tibble: 1 x 1  
##   corr_weight  
##   <dbl>  
## 1      0.865
```


Blood pressure plot before correlation

What about our blood pressure data from NHANES?



Time plots

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Correlation for blood pressure data

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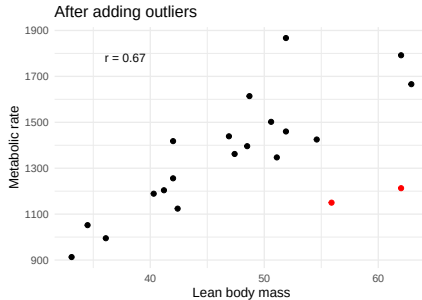
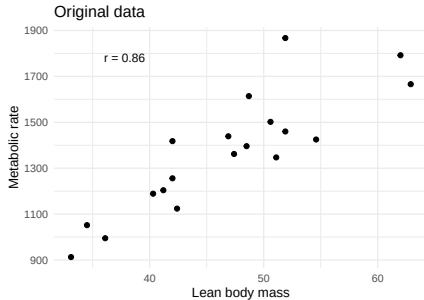
Assessing a relationship
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correlation

```
bp_cor <- nhanes_data %>%  
  summarize(corrbp = cor(bpxsy1, bpxdi1))  
bp_cor
```

```
## # A tibble: 1 x 1  
##   corrbp  
##   <dbl>  
## 1  0.322
```

Correlation is not resistant to outliers

Adding two unusual points changes the correlation for the weight data.



Time plots

Relationships between two
quantitative variables

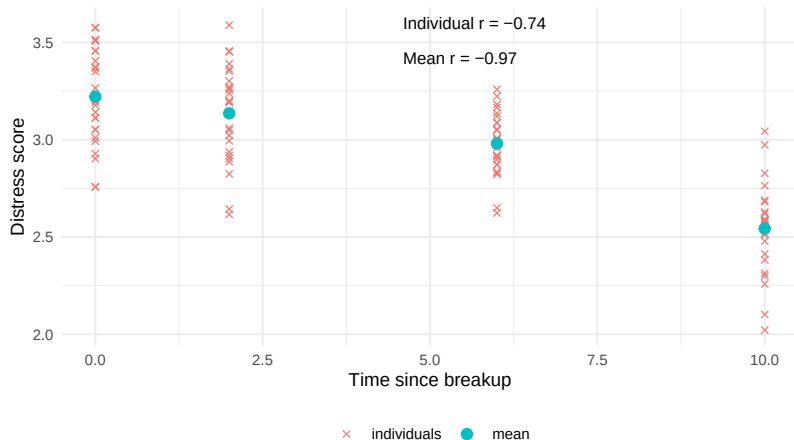
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Correlations for averages versus individuals

- Correlations based on averages are often stronger than correlations based on individual data.



Time plots

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Assessing a relationship
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- ▶ Determine which variable is explanatory and which is response, or when it doesn't matter
- ▶ Visually describe the relationship between two variables (form, direction, strength, and outliers)
- ▶ Numerically describe the relationship with the correlation coefficient r

Time plots

Relationships between two quantitative variables

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R Recap: What functions did we use?

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- ▶ `geom_point()`
- ▶ `geom_line()`
- ▶ `aes(col = gender)` to color points by levels of gender
- ▶ `summarize()` to calculate correlation using `cor(var1, var2)`

Reminder: Association does not equal causation

Time plots

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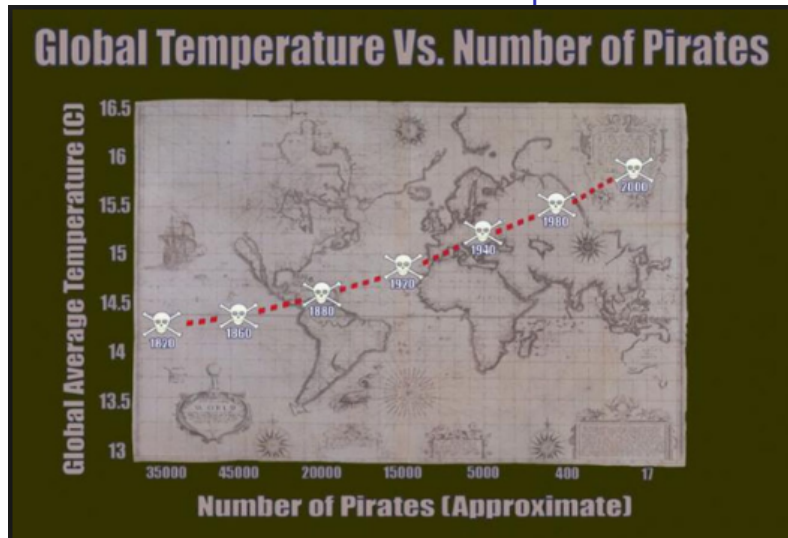
Assessing a relationship
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Remember that just because two variables are associated, does not mean there is a causal relationship

The correlation coefficient measures association *not* causation.

Even a very strong association doesn't mean that one variable causes the other.

Reminder: Association does not equal causation



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This image is one from a Forbes.com article but this example pops up in lots of places